Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	Claypot Chicken Rice
Project / Solution Name	Jobee
Name (Leader)	Lam Way Hou
Email & Phone (Leader)	lamwayhou@gmail.com 016-221-1236
University Affiliation(s)	Sunway University, Universiti Putra Malaysia
Live Prototype URL	https://jobee-sdchack.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/jimmerino2/sdc_hackathon_2026.git

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Users interact on a Vue 3 single-page app, hosted on Vercel and Replit. It calls a FastAPI backend on Replit over REST. The backend checks logins with Supabase Auth and stores profiles, roadmaps and Bonus Quest submissions in Supabase's Postgres database. For AI features, the backend calls the Grafilab LLM gateway, which runs Qwen3.7-Flash. The model writes learning roadmaps, generates and judges Bonus Quests, and extracts skills from uploaded resumes. Resume scoring can go to our self-hosted OpenJev pod first and falls back to the LLM if the pod is down. To judge a quest, the backend reads the submitted repository through the GitHub API.

Python pipeline scripts download public datasets: Indeed Hiring Lab, O*NET, DOSM, HN Algolia, FRED, Census BTOS, Epoch AI, METR and Stack Overflow. They clean and join the data. Reference tables such as roles and skills are loaded into Supabase. We trained our own forecasting models (LightGBM, ridge and trend models), partly on Runpod.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.14.5 (needs ≥3.12)	Backend, data pipeline, ML	PSF
TypeScript 5.9.3	Frontend	Apache-2.0
SQL / PostgreSQL (Supabase)	Database schema and migrations	PostgreSQL
Vue 3.5.44	Frontend UI	MIT
Vue Router 5.4.0	Page routing	MIT
Vite 7.3.7	Frontend dev server and build	MIT
FastAPI 0.157.0	Backend REST API	MIT
Uvicorn 0.54.0	Server that runs FastAPI	BSD-3
Pydantic 2.14.1 / pydantic-settings 2.16.1	Data validation and app config	MIT
supabase-py 2.33.0	Database and auth client	MIT
openai 3.35.1	LLM calls (roadmaps, quests, resumes)	Apache-2.0
httpx 0.28.1	HTTP calls to GitHub and OpenJev	BSD-3
pypdf 6.21.2	Reading uploaded resume PDFs	BSD-3
pandas 3.0.6 / NumPy 2.5.4	Data processing	BSD-3
LightGBM 4.7.0	Forecast model	MIT
scikit-learn 1.9.1 / SciPy 1.18.1	Modelling and stats helpers	BSD-3
openpyxl 3.1.5	Reading the DOSM Excel workbook	MIT
PyYAML 6.0.3	Reading METR benchmark data	MIT
lucide-vue-next 1.0.0	Icons	ISC
pytest 9.1.1 / Vitest ^4.1	Tests	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Supabase	Database, login, magic-link emails	Free
Grafilab LLM gateway	Roadmap, Bonus Quest and resume generation	Free
GitHub REST API v3	Reading submitted Bonus Quest repos	Free
OpenJev `/v1/systemone`	Resume parsing (falls back to the LLM)	Free
Google Fonts (Nunito)	Frontend and email typography	Free
Indeed Hiring Lab job_postings_tracker + ai-tracker (git repos)	Job-posting volume and AI posting share	Free
DOSM Job Market Insights Q2 2026 (xlsx)	Malaysian vacancies by occupation	Free
O*NET database 31.0	Occupations, skills, job zones	Free
HN Algolia Search API v1	"Who is hiring" ads (occupational training data)	Free
Adzuna Jobs API v1	Job-ad cross-check only	Free
Stack Overflow Developer Survey 2023–2025	Developer AI-tool usage	Free
FRED / ALFRED API	Real-Time Population Survey AI usage	Free
US Census BTOS	Business AI adoption	Free
Epoch AI benchmark data + all_ai_models.csv	AI capability and time-horizon data	Free

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Wayback Machine CDX + captures	Older dated copies of Epoch's data	Free
METR benchmark_results 1.1 + eval-analysis-public repo	AI task time horizons	Free
Anthropic Economic Index (release 2026-06-26, Hugging Face)	Task exposure to AI	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Host frontend & backend prototype	Sponsor-provided
Supabase	Cloud PostgreSQL database & Vector store	Free
RunPod	Dedicated GPU/Compute cloud used to host and run the automated Judge Agent for background evaluation and scoring task	Sponsor-provided

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Qwen3.7-Flash (`qwen/qwen3.7-flash`)	Alibaba Qwen, served through the Grafilab gateway	Pre-trained, used as-is: roadmaps, Bonus Quest writing and judging, resume skill extraction, chat	Qwen model license + Grafilab terms (not checked)
DeepSeek-V4-Flash (`deepseek-v4-flash`)	DeepSeek, through Grafilab or OpenRouter	Pre-trained: alternative model in `.env.example`	DeepSeek license + provider terms (not checked)
OpenJev (`openjev`, `/v1/systemone` endpoint)	Self-hosted pod	Pre-trained: scores which skills a resume shows	Not checked
LightGBM sector forecaster (4 models: h3, h6, h9, h12)	Self-trained (train_lgbm.py) on Indeed postings + AI exposure data	Self-trained: job-postings index forecasts per sector	Project's own (Apache-2.0); training data CC BY 4.0
Occupational skill-demand models (persistence, mean12, shrink, trend, ridge, blend)	Self-trained (occupational/models.py), scikit-learn Ridge on HN ads	Self-trained: skill-demand forecasts, chosen by backtest	Project's own (Apache-2.0); HN content under HN terms
Forecasting candidates (naive, drift, damped drift, pooled mean, local trend, ridge, LightGBM)	Self-trained (forecasting/models.py) on BTOS, RPS, Indeed data	Self-trained: AI adoption and demand forecasts	Project's own (Apache-2.0); data public domain / CC BY

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
AI capability engine (trend + logistic success curves, bootstrap)	Self-built (capability_engine.py) on Epoch + METR data	Self-trained: AI capability / time-horizon scenarios	Project's own (Apache-2.0); data CC BY 4.0

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Indeed Hiring Lab Job Postings Index	github.com/hiring-lab/job_postings_tracker	Public	CC BY 4.0	N
Indeed Hiring Lab AI Tracker	github.com/hiring-lab/ai-tracker	Public	CC BY 4.0	N
DOSM Job Market Insights Q2 2026	dosm.gov.my	Public	DOSM terms of use	N
O*NET Database 31.0	onetcenter.org	Public	CC BY 4.0	N
HN "Who is hiring?" posts	hn.algolia.com (news.ycombinator.com)	Public	HN terms	Y: stores poster usernames, and ad text can include contact names and emails
Adzuna job ads	api.adzuna.com	Public	Adzuna API terms	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Stack Overflow Developer Survey 2023–2025	survey.stackoverflow.co	Public	ODbL	N: anonymous, and only 4 columns are read
Real-Time Population Survey (GenAI use)	fred.stlouisfed.org / realtimepopulationsurvey.com	Public	See source	N: published aggregates only
Census BTOS	census.gov/hfp/btos	Public	Public domain	N
Epoch AI benchmark data + model list	epoch.ai/data	Public	CC BY 4.0	N
METR benchmark results 1.1 + eval-analysis-public	metr.org, github.com/METR	Public	Not checked	N
Anthropic Economic Index (2026-06-26)	huggingface.co/datasets/Anthropic/EconomicIndex	Public	CC BY	N: aggregated
Skill learning resources	Hand-picked links (0010_skill_sources.sql)	Public	Each site's own terms	N
App user data (profiles, resumes, roadmaps, quest submissions)	Supabase (your project)	Private	Your own data, no license	Y: name, email, skills, work experience, uploaded resume text, GitHub repo URLs

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code (Claude Opus 5.5, some Claude Sonnet 5.5)	Backend / Frontend	Major
Replit Agent	Frontend	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Every AI-generated change went through a team member before it reached main. We read the diffs, ran the code, and merged through feature branches. We ran the app locally and on Replit to check the endpoints and pages by hand. We held forecasts to stricter checks: rolling-origin backtests, held-out test periods, and hyperparameters fixed before testing so results couldn't be tuned to look good. We also kept the AI forecast branch out of main until it had passed review. AI-assisted commits are labelled as co-authored, and we, not the tools, are responsible for what we shipped.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Cloud Platform	Hosting and deploying the live prototype
RunPod	Cloud GPU Compute	Hosting the automated Judge Agent to evaluate matching output

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit was essential for rapid deployment and accessibility of our prototype without local environment configuration. RunPod provided the necessary dedicated compute to run our automated Judge Agent in the background. While the core logic could be migrated, these sponsors' seamless integration was crucial for demonstrating a complete, production-ready proof-of-concept during this phase

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Jobee processes user-submitted resumes and job queries for matching purposes. We do not use personal identifiable information (PII) for model training. All API keys and tokens (OPENROUTER_API_KEY, OPENJEV_TOKEN) are securely stored in environment variables and are not exposed in the repository. User data is securely stored in Supabase, protected by Row Level Security (RLS) policies. Data sent to the RunPod Judge Agent is strictly for real-time evaluation and scoring, and is not used for external training. All third-party API calls are made via encrypted HTTPS connections.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The main technical limitation is the dependency on third-party APIs (OpenRouter, OpenJev, RunPod). Network latency or rate limits from these providers could affect response times. Additionally, LLMs can occasionally hallucinate or produce biased recommendations. To mitigate these risks, we implemented error-handling and retry mechanisms in our FastAPI backend. We also set strict system prompts to constrain the AI's output and ensure factual accuracy. Finally, we track the system's F1-score to monitor matching quality continuously.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- [x] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- [x] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- [x] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- [x] Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- [x] Any use of official SDC sponsor resources has been declared in Section 5.
- [x] The prototype described here is accessible to judges through the links provided in Section 1.
- [x] All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



25/10/2026